## 1.3: Rates of Change and Behavior of Graphs

**Definition. (Rate of Change)**
A rate of change describes how an output quantity changes relative to the change in the input quantity. The units on a rate of change are "output units per input units". The **average rate of change** between two input values is the total change of the function values divided by the input values.

$$\frac{\Delta y}{\Delta x} = \frac{f(x_2) - f(x_1)}{x_2 - x_1}$$

**Example.** Suppose the following table contains temperatures recorded through the day.

| $h$ | 7 AM | 9 AM | 11 AM | 1 PM | 3 PM |
|---|---|---|---|---|---|
| $T(h)$ | $78°F$ | $81°F$ | $87°F$ | $93°F$ | $98°F$ |

Find the average rate of change between

7 AM and 9 AM

$$\frac{81-78}{9-7} = \boxed{\frac{3}{2}\ °F/hr}$$

9 AM and 11 AM

$$\frac{87-81}{11-9} = \frac{6}{2}\ °F/hr = \boxed{3\ °F/hr}$$

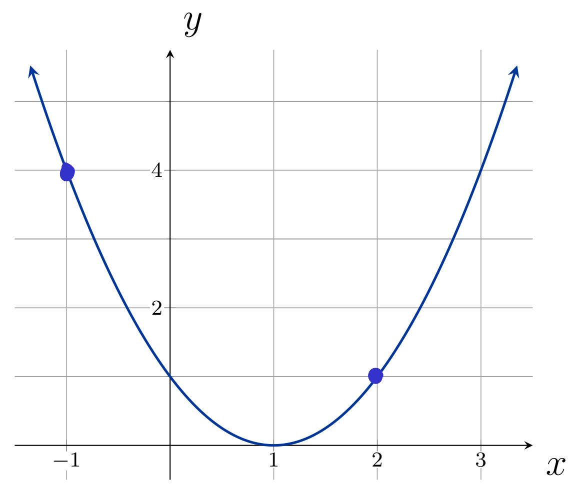
15:00

7 AM and 3 PM

$$\frac{98-78}{15-7} = \frac{20}{8}\ °F/hr = \boxed{\frac{5}{2}\ °F/hr}$$

**Example.** Using the following graph, determine the average rate of change on the interval $[-1, 2]$

$$\frac{f(2) - f(-1)}{2 - (-1)} = \frac{1 - 4}{2 + 1} = \frac{-3}{3} = -1$$




**Example.** Find the average rate of change of $f(x) = x^2 - \dfrac{1}{x}$ on the interval $[2, 4]$.

$$\frac{[f(4)] - [f(2)]}{4 - 2} = \frac{\left[4^2 - \frac{1}{4}\right] - \left[2^2 - \frac{1}{2}\right]}{2}$$

$$= \frac{16 - \frac{1}{4} - 4 + \frac{1}{2}}{2}$$

$$= \frac{12.25}{2}$$

$$= 6.125 = \frac{49}{8}$$

**Example.** Find the expression for the average rate of change of $g(t) = t^2 + 3t + 1$ on the interval $[0, a]$.

$$\frac{g(t) - g(a)}{t - a} = \frac{[t^2 + 3t + 1] - [a^2 + 3a + 1]}{t - a} = \frac{t^2 + 3t + 1 - a^2 - 3a - 1}{t - a}$$

$$= \frac{(t^2 - a^2) + 3t - 3a}{t - a} = \frac{\cancel{(t-a)}(t+a) + 3\cancel{(t-a)}}{\cancel{t-a}} = \boxed{t + a + 3}$$

**Example.** Find the expression for the average rate of change of $j(t) = \dfrac{1}{t+3}$ on the interval $[1, 1+h]$.

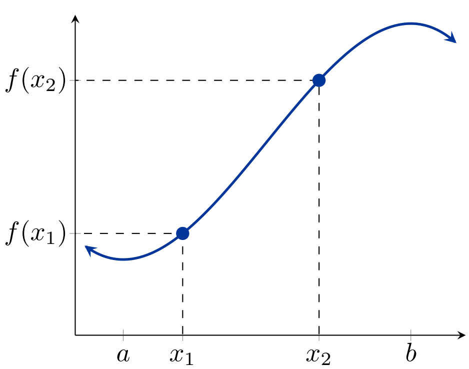
$$\frac{j(1+h)-j(1)}{(1+h)-1} = \frac{\frac{1}{1+h+3} - \frac{1}{1+3}}{h} = \frac{\left(\frac{4}{4}\right)\frac{1}{4+h} - \frac{1}{4}\left(\frac{4+h}{4+h}\right)}{\frac{h}{1}} = \frac{\cancel{4} - (\cancel{4}+h)}{h \cdot 4(4+h)}$$

$$= \frac{-\cancel{h}}{4\cancel{h}(4+h)}$$

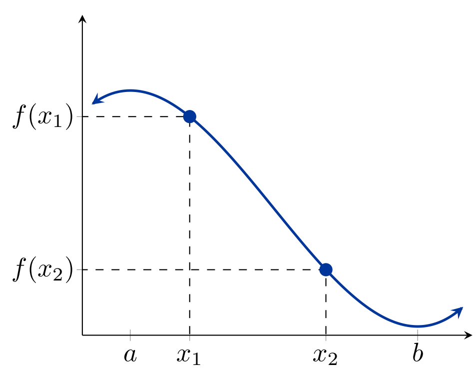
$$= \boxed{\frac{-1}{4(4+h)}, \quad h \neq 0}$$

**Definition.**
Consider the function $f(x)$ on the interval $(a, b)$. Given *any* two numbers $x_1$ and $x_2$ in $(a, b)$ where $x_1 < x_2$, we say $f$ is

increasing if $f(x_1) < f(x_2)$

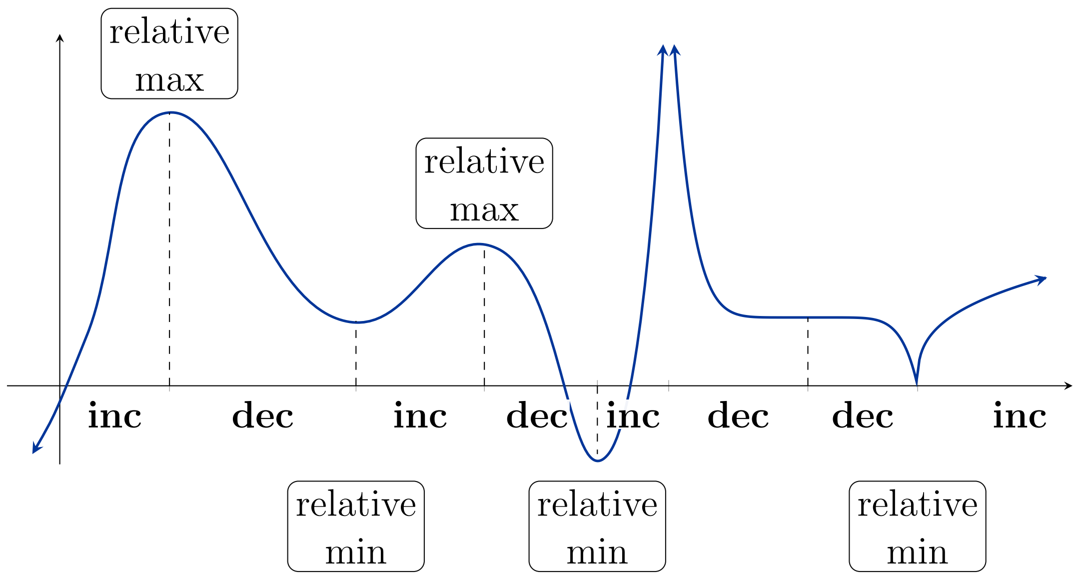



decreasing if $f(x_1) > f(x_2)$



**Definition. (Relative/Local Extrema)**
A function $f$ has a

- **relative maximum** at $x = c$ if $f(c) \geq f(x)$ for every $x$ in $(a, b)$
- **relative minimum** at $x = c$ if $f(c) \leq f(x)$ for every $x$ in $(a, b)$




**Definition. (Absolute Extrema)**
A function $f$ defined on a domain $D$ has an

- **absolute maximum** at $x = c$ if $f(c) \geq f(x)$ for every $x$ in $D$
- **absolute minimum** at $x = c$ if $f(c) \leq f(x)$ for every $x$ in $D$

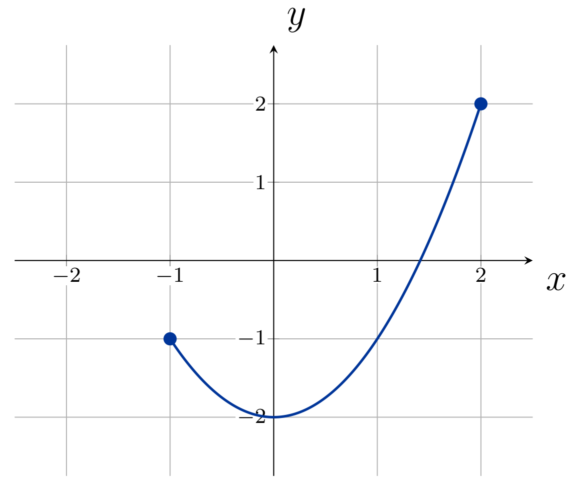
**Example.** Using the following graph, find any relative and absolute extrema.

rel. min of $-2$ at $x=0$

No rel. max

Abs. min of $-2$ at $x=0$

Abs. max of $2$ at $x=2$



**Example.** Using the following graph, find any relative and absolute extrema.

rel. min of $0$ at $x=0$

rel. max of $16$ at $x=\pm 2$

Abs. min of $-9$ at $x=3$

Abs. max of $16$ at $x=\pm 2$